Xiyuan Wang website *Resume Updated on: February 2, 2025*

Contact Information

School of Information Science and Technology
ShanghaiTech University
393 Middle Huaxia Road, Pudong, Shanghai, China

@ wangxy7@shanghaitech.edu.cn
horanny.github.io
(+86) 136 3639 4580

Research Interests

Human-Computer Interaction (HCI), Game Storytelling, Game Design, Creativity Support
Human-AI Collaboration, Data Visualization

Education

ShanghaiTech University, Shanghai, China

Master of Computer Science — Sept. 2022 - Jun. 2025

Advisor: Quan Li

ShanghaiTech University, Shanghai, China

B.Eng. in Computer Science — Sept. 2018 - Jun. 2022

Advisor: Quan Li

Academic Experience

ViSeer Lab, ShanghaiTech University, Shanghai, China

Graduate Research Assistant (Advisor: Quan Li) — 2022 - present

Research focused on visual analytics systems for game algorithms and creativity support tools for game storytelling. My work includes:

- Designed and evaluated a visual analytics framework integrating a scalable churn prediction model with two eXplainable AI methods, providing actionable insights to game designers to reduce player churn. Conducted user interviews and a within-subjects study to validate its effectiveness and usability in real-world scenarios.
- Developed and evaluated a visual analytics framework for in-game friend recommendation systems, incorporating methods to balance similarity and diversity. This framework supports fine-tuning recommendation processes based on player feedback and behavior.
- Created and evaluated an in-game mod for automatic clue capture and classification, enabling interactive storytelling through an open-source platform. Conducted a between-subjects study to evaluate the effectiveness and usability of the system.

The HCI Initiative, Hong Kong University of Science and Technology, Hong Kong, China

Graduate Research Assistant (Advisors: Xiaojuan Ma, Jianping Gan) — 2023 - 2024

Visual analytics on South China Sea.

- Processed 30 days of 3D ocean data, generating vector fields for ocean currents and performing ocean volume modeling to create a realistic environmental dataset for analysis.
- Built an interactive visual analytics system in Unity 3D, integrating Cinema4D for terrain modeling, Unity VFX Graph for ocean currents, custom shaders for volume rendering, and interactions for precise ocean area localization, enabling users to explore and interact with the ocean environment.

User eXperience Center, NetEase, Shanghai, China

Research Internship (Advisor: Wei Wan) — 2022 - 2023 (Overlap with ViSeer Lab)

Contributed to the development of visual analytics frameworks for churn prediction and diverse friend recommendation systems, in collaboration with ViSeer Lab.

ViSeer Lab, ShanghaiTech University, Shanghai, China

Undergraduate Research Assistant (Advisor: Quan Li) — 2021 - 2022

Visual analytics systems for e-commerce.

Publications

Xiyuan Wang, Yifan Cao, Junjie Xiong, Sizhe Chen, Wenxuan Li, Junjie Zhang, Quan Li. ClueCart: Supporting Game Story Interpretation and Narrative Inference from Fragmented Clues. In *ACM CHI Conference on Human Factors in Computing Systems*.
Conditionally Accepted by CHI 2025 · Full Paper

Yuanhao Zhang, Yumeng Wang, **Xiyuan Wang**, Changyang He, Chenliang Huang, Xiaojuan Ma. CoKnowledge: Supporting Assimilation of Time-synced Collective Knowledge in Online Science Videos. In *ACM CHI Conference on Human Factors in Computing Systems*.
Conditionally Accepted by CHI 2025 · Full Paper

Xiyuan Wang, Ziang Li, Sizhe Chen, Wei Wan, Xingxing Xing, Quan Li. Incorporating Visual Analytics for Preference Mediation in Games: Balancing Similarity and Diversity in Friend Recommendations. In *ACM Conference on Intelligent User Interfaces*.
Conditionally Accepted by IUI 2025 · Full Paper

Quan Li, Yun Tian, **Xiyuan Wang**, Laixin Xie, Dandan Lin, Lingling Yi, Xiaojuan Ma. MetapathVis: Inspecting the Effect of Metapath in Heterogeneous Network Embedding via Visual Analytics. In *Computer Graphics Forum*.
CGF 25 · Full Paper

Xiyuan Wang*, Laixin Xie*, He Wang, Xingxing Xing, Wei Wan, Ziming Wu, Xiaojuan Ma, Quan Li. Deciphering Explicit and Implicit Features for Reliable, Interpretable, and Actionable User Churn Prediction in Online Video Games. In *IEEE Transactions on Visualization and Computer Graphics*.
IEEE TVCG 24 · Full Paper

Longfei Chen, Chen Cheng, He Wang, **Xiyuan Wang**, Yun Tian, Xuanwu Yue, Wong Kam-Kwai, Suting Hong, Quan Li. FMLens: Towards Better Scaffolding the Process of Fund Manager Selection in Fund Investments. In *IEEE Transactions on Visualization and Computer Graphics*.
IEEE TVCG 24 · Full Paper

Chenyang Zhang, **Xiyuan Wang**, Chuyi Zhao, Yijing Ren, Tianyu Zhang, Zhenhui Peng, Xiaomeng Fan, Xiaojuan Ma, Quan Li. PromotionLens: Inspecting Promotion Strategies of Online E-commerce via Visual Analytics. In *IEEE Transactions on Visualization and Computer Graphics (Proc. VIS 2022)*.
VIS 2022 · Full Paper

Papers in Preparation

Self-Learning Enhancement via Epistemologically-Informed LLM Dialogue
Under Review · Full Paper

Conference Presentations

A Visual Analytics Approach to Exploring the Feature and Label Space Based on Semi-structured Electronic Medical Records.
IEEE VIS 2023, Melbourne, Australia.

PromotionLens: Inspecting Promotion Strategies of Online E-commerce via Visual Analytics.
ChinaVis 2022, Xining, China.

Honors and Awards

Third Price of ChinaVis Data Challenge 2023
Second Price of ChinaVis Data Challenge 2022

Services

CHI; CHI lbw; Chinese CHI, Conference
Peer Review 2022 - 2025

ARTS 1422 - Data Visualization, ShanghaiTech University, China
Teaching Assistant Spring 2023

Kedu No. 1 Primary School, Qiannan Buyi and Miao Autonomous Prefecture, China
Volunteer Teacher 2019

Shanghai Marathon, Shanghai, China
Volunteer 2018, 2020

SKILLS

Research: Data Visualization, Human-Computer Interaction, Quantitative & Qualitative Research, Interview, Iterative Design.

Program Language: JavaScript, Python, C Sharp, SQL.

Framework and Tools: D3, Vue, MongoDB, PyTorch, Unity 3D, Cinema 4D, Figma.

Languages: Mandarin Chinese (Native), English (IELTS: 6.5, DET: 130), Japanese (Beginner), Visualization, LaTeX.